

Lecture 13, Feb 23-2026.

7.4 Integration of Rational Functions by Partial Fractions.

Notice that

$$\begin{aligned}\frac{x+5}{x^2+x-2} &= \frac{2}{x-1} - \frac{1}{x+2} \stackrel{?}{=} \frac{2(x+2) - (x-1)}{(x-1)(x+2)} \\ &= \frac{2x+4-x+1}{x^2+x-2} = \frac{x+5}{x^2+x-2}\end{aligned}$$

$$\begin{aligned}\text{But } \int \frac{x+5}{x^2+x-2} dx &= \int \left\{ \frac{2}{x-1} - \frac{1}{x+2} \right\} dx \\ &= \int \frac{2}{x-1} dx - \int \frac{1}{x+2} dx \\ &= 2 \ln|x-1| - \ln|x+2| + C \\ &= \ln \left| \frac{(x-1)^2}{x+2} \right| + C.\end{aligned}$$

* The Method of Partial Fractions.

Let $f(x) = \frac{P(x)}{Q(x)}$ where P, Q are polynomials.

Case 1: If $\deg(P) \geq \deg(Q)$ (f is improper) then, we can use division of polynomials (by long division) we have

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

$$\begin{array}{r|l} P(x) & Q(x) \\ \hline R(x) & S(x) \end{array}$$

Case 2: If $\deg(P) < \deg(Q)$ (f is proper) then this is a rational function.

Example Find $\int \frac{x^3+x}{x-1} dx$.

Sln. Notice that $\deg(x^3+x)=3 > \deg(x-1)=1$

$$\begin{array}{r} x^3+x \quad | \quad x-1 \\ \underline{-x^3+x^2} \quad x^2+x+2 \\ x^2+x \\ \underline{-x^2+x} \\ 0+2x \\ \underline{-2x+2} \\ 2 \end{array}$$

then, $\frac{x^3-x}{x-1} = x^2+x+2 + \frac{2}{x-1}$

$$\begin{aligned} \int \frac{x^3-x}{x-1} dx &= \int (x^2+x+2) dx + \int \frac{2}{x-1} dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + \ln(x-1)^2 + C. \end{aligned}$$

If the denominator $Q(x)$ is factorable, then:

$$Q(x) = (x^2-4)(x^2+4) = (x-2)(x+2)(x^2+4)$$

$$x^2+4, \quad a=1, \quad b=0, \quad c=4.$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}; \quad b^2-4ac \geq 0$$
$$0^2 - 4(1)(4) = -16 < 0$$

then, we can write the linear polynomials as

$$\frac{A}{(ax+b)}$$

and

$$\frac{Ax+B}{(ax^2+bx+c)}$$

then

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \dots + \frac{A_k}{a_kx+b_k} + \frac{B_1}{(c_1x^2+d_1x+e_1)} + \dots +$$

Example. Find $\int \frac{x^2+2x-1}{2x^3+2x^2-2x} dx$.

Sln.

$$2x^3+2x^2-2x = 2x(x^2+x-1) = 2x(x-\frac{1}{2})(x+2).$$

$$\frac{x^2+2x-1}{2x^3+2x^2-2x} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2}.$$

$$= \frac{A(2x-1)(x+2) + Bx(x+2) + Cx(2x-1)}{x(2x-1)(x+2)}$$

if $x=0$ then $A(0-1)(0+2) = 0^2+2(0)-1$

$$-2A = -1 \Rightarrow A = \frac{1}{2}.$$

if $x=-2$ then $2C(4-1) = (-2)^2-4-1 =$

$$6C = -1 \Rightarrow C = -\frac{1}{6}$$